

Traditional and Local Knowledge Survey in the Kluane Area and Identification of Upwelling Groundwater Areas in Kluane Lake

**The Yukon River Panel
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**Prepared by:
Jan McKenzie and Jane Wilson**

**Kluane First Nation
Box 20
Burwash Landing, Yukon
YOB 1V0**
*Phone: (867) 841-5501
Fax: (867) 841-5900
E-mail: KluaneFN@YT.sympatico.ca*

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ABSTRACT

This report summarizes information obtained on salmon resources in the Kluane area watersheds of the White River sub-basin from interviews with Kluane First Nation members and local residents. Emphasis was placed on identifying areas of upwelling groundwater in Kluane Lake to help identify potential chum spawning habitat. A total of 12 First Nation and five non-First Nation individuals were interviewed during April and May, 2003. Several areas in Kluane Lake were identified where upwelling groundwater was observed or possibly indicated by areas remaining open in the winter. These areas included: near the mouths of Spring, Silver and No Name Creeks, Brooks Arm (Little Arm) near the narrows, Thorsen Bay, Sandspit Point (Poison Point), Dutchman's Point, Cultus Bay/Creek, Christmas Creek and at the mouths of Copper Joe Creek, Raft Creek and the Gladstone River. Other information was collected on the freshwater fishery for Kluane First Nation. This is not summarized in the report but is available for the information of Kluane First Nation on interview transcripts and tapes.

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1.0 INTRODUCTION

This report summarizes the findings on salmon resources from both traditional and local knowledge in the Kluane area watersheds in the White River sub-basin. It is important to identify areas of upwelling groundwater in the Kluane Lake area to help identify potential chum spawning habitat. Other information was collected on the freshwater fishery for Kluane First Nation. Those findings are not reported here but are available from interview transcripts and cassette recordings on file at the Kluane First Nation, Lands and Resources office, Burwash Landing.

The main objectives of this project were to:

1. Compile information on the salmon resource found within the Kluane First Nation traditional territory through the collection of traditional and local knowledge;
2. Identify community concerns and solicit input into the community fisheries stewardship plan; and
3. Identify and map areas of upwelling ground water in Kluane Lake indicating potential chum spawning habitat.

2.0 METHODS

Through consultation with Kluane First Nation, 17 Kluane First Nation Elders and other members and 6 local non-First Nation individuals were identified and contacted as individuals with possible information on the salmon resource in the Kluane area. Of these, 12 First Nation and 5 non-First Nation individuals agreed to be interviewed. Others declined interviews because of health reasons, because they were not available, or that they felt they had nothing to contribute. The findings from the interviews are summarized in table form in Appendix 4.

All interviews were conducted in their homes or at a place of convenience by two fishery technicians. Interviews were conducted over a 12-day period in April and May 2003. All interviews were taped on a cassette recorder except for two individuals who preferred not to be taped. These interviews were recorded by hand in note form. Photographs were taken by permission of the individual being interviewed. The interviews ranged from 1.5 to 3.5 hours. Cassette tapes were transcribed when possible. The tape of Dennis Dickson was not transcribed but relevant salmon information was recorded on paper while being listened to. There are some gaps in the recordings, when the sound is inaudible due to poor quality, or when it is difficult to understand quiet voices.

Questions were based on knowledge of the salmon resource in specific watersheds of the Kluane First Nation traditional territory, including: the Kluane River, Kluane Lake, the Nisling River, Tincup Creek/Lake, the Donjek and Koidern Rivers. Topographic maps of the watersheds, both 1:250,000 and 1:50,000 were used during the interviews to help with the identification of salmon spawning grounds, areas of harvest, areas of upwelling groundwater and general salmon observations. Questions were different for traditional knowledge interviews and for local knowledge interviews (see questions, Appendix 1&2).

The interview questions do not attempt to capture all information about the salmon fishery of Kluane First Nation. Previous reports by Horler (1994) and 'A Traditional Knowledge Survey On Salmon, by Kluane First Nation' (1998) document and describe the salmon fishery in great detail. The interviews for this project were designed to capture information gaps in previous reports, particularly on areas of open water, upwelling groundwater and possible chum spawning habitat.

Information on the freshwater fishery was also collected but not summarized in this report. This included information on spawning areas, abundance and distribution, historical information and management concerns about lake trout, burbot, pike, grayling, suckers and whitefish.

3.0 RESULTS - Summary of Responses

All locations referred to in this report are indicated on accompanying maps (Figs. 1, 2a, 2b). The major watersheds and salmon bearing streams of the White River sub-basin lying within the traditional territory of Kluane First Nation are shown in Figure 1. Figures 2a and 2b show identified areas of winter open water and salmon catches or observations on Kluane Lake. In some instances, specific locations were referred to by one or more names. To avoid confusion, all local names are shown in parenthesis.

3.1 Traditional Knowledge of the Salmon Resource

The presence of salmon was reported in the following watersheds: Kluane Lake (Luan Man), Tincup Lake/Creek (Gyu Chu), and the Kluane, Donjek, Koidern and Nisling Rivers. Interviewees could not always specify whether the salmon identified was chum or chinook. Historically and today, the most well known chum spawning area as identified in the interviews was 'Salmon Patch' (The Ku) located on the Kluane River. Other chum spawning areas identified on the Kluane River were Swede Johnson and Glacier Creek (Mile 1118) Sloughs. The sites identified on Kluane Lake as possible chum spawning areas were, Cultus Bay, Christmas Bay, Dutch Harbour and near the mouths of the Gladstone River and Copper Joe Creek (Half Breed Creek)¹. Historically, chum have been netted periodically on Kluane Lake near Jackson Point (Bayshore), Cultus Bay, Burwash Point, the mouth of Silver Creek and the lake outlet (Down Below). Spawning chum salmon have also been observed on the Koidern River.

Tincup Lake and Creek were identified as of great historical importance for chinook salmon. Several individuals had family camps there a long time ago and reported having seen old fish traps in the area. Historically, chinook salmon have been observed or caught in nets on Kluane Lake at the following locations: the mouths of Copper Joe and Spring Creeks, near Burwash Point and Burwash Landing and up Brooks Arm (Little Arm). Chinook have also been observed near the mouths of the Donjek and Nisling Rivers.

¹ This creek was previously named Halfbreed Creek and recently renamed Copper Joe Creek.

Areas of Kluane Lake that were identified as possible locations of upwelling groundwater included: Brooks Arm at the narrows, Thorsen Bay, Sandspit Point (Poison Point), Dutchman's Point, Cultus Bay, near the mouths of Spring, Silver, No Name, Christmas, Copper Joe and Raft Creeks and the Gladstone River.

Chum salmon is rarely consumed by humans for food but is still used occasionally for trapping bait or dog food. Gaffing and clubbing are the most common ways of harvesting chum. Chinook salmon is not currently harvested in the traditional territory.

3.2 Local Knowledge of the Salmon Resource

Chum salmon have been observed or harvested in the following Kluane River locations: Glacier Creek (Mile 1118) Slough, Swede Johnson Slough and 'Salmon Patch'. Incidental catches of chum have been recorded in Kluane Lake. Another chum spawning area is Christmas Creek on Kluane Lake.

Chinook spawning areas were identified on the Nisling River in the lower reaches and at the mouth of Onion Creek. Chinook have been observed at the mouth of the Donjek River and on the Klotassin River above the confluence of Dip Creek. No one interviewed had knowledge of chinook salmon being caught in nets in Kluane Lake.

Areas of open water or groundwater were identified at the following locations: Brooks Arm just before the narrows, Cultus Bay, Thorsen Bay and near the mouth of Raft Creek.

4.0 RECOMMENDATIONS- by interviewees

There were several recommendations regarding the freshwater fishery, but few for the salmon fishery. The main salmon concern was regarding beaver dams on the Kluane River and how they may impact the spawning chum salmon habitat. Several First Nation members recommended these dams be removed. There was also concern raised by a local resident about the use of ATV's at creek crossings where salmon spawn. It was suggested that signage be posted by spawning channels or streams.

5.0 CONCLUSIONS

There is one documented chum spawning area in Kluane Lake that is located near the mouth of Silver Creek (Wilson, 2004). However, there are likely chum spawning areas at several other locations along the east and south side of Kluane Lake especially near Christmas Creek and Cultus Bay. Chum can be caught along the west side of the lake, including Goose Bay and Dutch Harbour, but spawning grounds are unknown. Although the presence of chum is documented in many areas of Kluane Lake, numbers are difficult to determine. Small runs are difficult to monitor in such a large geographical area. As well, timing of chum surveys is crucial and weather conditions can be challenging in the fall when chum are spawning. Most of the individuals interviewed had little knowledge or only "hearsay" knowledge of chum spawning on Kluane Lake. Many fishers had more interest in freshwater fish since they are easily accessible and fairly abundant. Chum salmon is used sparingly for trap bait and dog

food and only occasionally for human food. Kluane First Nation members do utilize Glacier Creek Slough (Mile 1118) or 'Salmon Patch' if they need chum.

The majority of individuals interviewed had little knowledge of chinook activity in Kluane Lake or the other watersheds. Sightings or knowledge of chinook outside the Tincup area were limited.

6.0 Acknowledgments

The authors wish to extend their sincere thanks to the people in Burwash Landing and Kluane area who were interviewed for giving their time and sharing their knowledge and stories. Thanks also to Grace Cohoe, of Kluane First Nation, for her valuable assistance in identifying individuals and setting up interviews, and Brad Wilson for assisting with interviews. Special thanks to Kim Sparks for transcribing tape recordings. Also, thanks to Luke Johnson and Sonia Hansen for preparing maps and Al von Finster for providing helpful comments on this report.

7.0 Literature Cited:

Horler, Aileen. 1994. A Review of the Salmon Resource found within the Traditional Territory of the Kluane First Nation. Unpublished report prepared for Kluane First Nation.

Compilation of transcribed interviews conducted in Burwash Landing October-December, 1998--Traditional Knowledge Survey. Prepared for Kluane First Nation.

Wilson, J. 2004. A Preliminary Investigation of Chum Salmon Spawning in Kluane Lake. R&E project 57-02 prepared for the Yukon River Panel.

Appendix 1. General Questions for Kluane First Nation Traditional Knowledge

Traditional and Local Knowledge Survey in the Kluane Area and Identification of Upwelling Groundwater Areas in Kluane Lake

1. Age or year born?
2. How long have you lived in the Kluane Area?
3. Observations or knowledge of freshwater fish and salmon on a drainage basis including: Kluane River, Kluane Lake, Nisling River, Tincup Creek/Lake, Donjek, Koidern Rivers.
 - Where did you travel or spend time in the Kluane Area?
 - What did you fish for?
 - Numbers of fish observed or harvested?
 - Location of spawning areas?
4. What methods did you use for fishing?
5. Do you fish now? What kind of fish?
6. Stories about fishing that have been passed down.
7. Have you noticed changes in fish numbers?
8. Have you observed spawning chum on Kluane Lake?
9. Have you observed open water areas in the winter/upwelling groundwater on Kluane Lake?
10. What do you think can be done to increase and/or protect the salmon resource?

Appendix 2. General Questions for Local Knowledge

Traditional and Local Knowledge Survey in the Kluane Area and Identification of Upwelling Groundwater Areas in Kluane Lake

1. How long have you lived in the Kluane Area?
2. Observations of fish, both freshwater and salmon throughout the Kluane area. Where?
When?
3. What spawning areas do you know of?
4. Have you observed open water in winter or upwelling groundwater areas on Kluane Lake?
5. Have you observed declines in catches?
6. Have you observed salmon fry in fish stomachs? Fish abnormalities?
7. Have you observed tagged fish?
8. Do you have any management concerns or suggestions?

Appendix 3 List of KFN Members and Kluane Area Residents Interviewed

Traditional and Local Knowledge Survey in the Kluane Area and Identification of Upwelling Groundwater Areas in Kluane Lake

Kluane First Nation Traditional Knowledge- Burwash Landing

Interviewed

Joe Johnson
Peter Johnson
Sharon Kabanak
Kirk Johnson
Grace Johnson
Agnes Johnson
Lena Johnson
Mary Easterson
Ron Chambers
Dorothy Johnson
Dennis Dickson
Alexander Dickson

Not interviewed

Grace Chambers-not feeling up to an interview
Josie Sias- declined
Kluane Martin-schedule differences

Local Knowledge- Burwash Landing, Destruction Bay, Cottonwood Park

Interviewed

Ollie Wirth
Ken Reader
Dan Washing
Gary Vanderveen
Mary Ann and Glenn

Not interviewed

John Ostashek- health issues

Appendix 4 Summary Tables of Information Collected.

The following tables provide summarized information about salmon obtained from transcriptions of taped interviews in 2002 as well as observations of winter open water and upwelling groundwater areas in Kluane Lake. Individual transcripts are available on file at the Kluane First Nation, Lands and Resources office in Burwash Landing.

Agnes Johnson Burwash Landing	Koidern River	Tincup Lake/Creek	Donjek	Kluane River	Kluane Lake
Spawning areas				Glacier Creek, 1118 [p.3]	
Salmon observation/ Fishing sites		Chinook salmon [p.4]		Harvest chum salmon at 1118 [p.3] Harvest chum at Swede Johnson [p.3] Dried chum for dog food [p.4]	Chinook salmon Harvested in the 70's by Grace Chambers [p.2] Net nr Silver Creek and Bayshore for Chum salmon in fall [p.9]
Fishing Methods		Fish traps at Tincup [p. 11]	Fish traps [p.11]	Nets [p.3, p.9] Clubs, sticks [p.4] Fish traps [p.11]	

Agnes Johnson	Koidern R	Tincup Lake/Creek	Donjek	Kluane River	Kluane Lake
Fishing stories	“I hear stories of people fishing up there. They say they get salmon.” [p.8]	“When we were in Tincup when we were young we used to see whole bunch of salmon, lake would be red with it. With king salmon.” [p.4]		Chum at Swede Johnson...”We would get so many salmon and come up and just pull them out by our hands. There were so many of them! We just clubbed them with sticks and pull them out with our hands and everything like that. We went crazy down there with the salmon.” [p.4]	
Recommendations				Beaver dams should be removed at 1118 and Swede Johnson for spawning salmon [p.10]	

Alexander Dickson Burwash Landing	Nisling River	Tincup Lake/Creek	Kluane River	Kluane Lake
Spawning areas				
Salmon observation/ Fishing sites	Just before it runs into the Donjek (salmon observed from Helicopter) [p.2]	Chinook [p.1] Chinook observed nr camp at the end of the Lake [p.3]		Chinook caught with parents [p.1]
Fishing Methods				
Areas of open water/ Spring water			Spring Creek open in winter [p.3]	Cultus Bay [p.4]

Dan Washing Resident 10 Yrs Burwash Landing	Kluane River	Kluane Lake
Spawning areas		
Salmon observation/ Fishing sites	Dog food chum [p.1] Mile 1118, Glacier Creek Slough [p.2]	
Areas of open water/ Spring water		
Fishing stories		I've heard of people catching (salmon) but I think Mike Williams...he had caught some." [p.2] "I haven't seen any, haven't snagged anything, nothing in the net (salmon) [p.3]

Dennis Dickson Burwash landing	Koidern River	Donjek	Kluane River	Kluane Lake
Spawning areas	Spawning chum observed	Chinook near the mouth	Salmon patch	Caught chum in net near Bayshore/October Heard chinook salmon up the Little Arm No knowledge of other Chinook in Lake
Salmon observation/ Fishing sites		Chinook near the mouth		
Areas of open water/ Spring water				Little Arm at narrows Thorsen Bay Poison Point Dutchman's Point Spring Creek

Dorothy Johnson Burwash Landing	Tincup Lake/Creek	Kluane River	Kluane Lake
Spawning areas			Chinook salmon had been seen spawning near Burwash Pt. Dog salmon spawning at Christmas Bay
Salmon observation/ Fishing sites	Used to be a lot of chinook salmon. People used to walk there and camp for a length of time to dry fish.	Lots of salmon at salmon patch (Sam Johnson her grandfather) and family had a camp there. Swede Johnson area for salmon	

Gary VanDerveen Resident 47yrs Destruction Bay	Kluane River	Kluane Lake
Salmon observation/ Fishing sites	Chum mile 1118 [p.1]	
Areas of open water/ Spring water		Little Arm just before the narrows [p.3]
Fishing stories		No knowledge of salmon being caught on lake

Grace Johnson Burwash Landing	Tincup Lake/Creek	Kluane River	Kluane Lake
Spawning areas			
Salmon observation/ Fishing sites	Chinook salmon in Tincup Creek [p.1]	Chum [p.1] Glacier Creek [p.2] Salmon Patch [p.3]	Harvest chum at Burwash Point [p.1]

Joe Johnson Burwash Landing	Nisling River	Koidern River	Tincup Lake/Creek	Kluane River	Kluane Lake
Spawning areas					Chinook? Spawn near the outlet [p.8 #1]
Salmon observations/ Fishing sites		Salmon in the mouth [p.10 #1]	Chinook salmon observed [p.5 #1]	Chum at salmon patch across from 1118 [p.7 #1]	Salmon at Dutch Harbour [p.4, 9 #1] Spring Creek [p.4 #1] Chinook nr. Burwash Landing [p.7 #1] 4 th of July [p.12 #2] Jarvis Ck. [p.13 #2]
Fishing Methods			Fish traps [p.5 #1] Historical fish traps nr mouth of lake [p.6 #2 p.8 #2]		Set nets [p.7 #1]
Areas of open water/spring water				Salmon Patch [p.7 #1]	Between Silver City and Christmas Bay [p.8 #1] Gladstone River runs year round [p.8 #1]

Ken Reader Resident Nr Burwash Landing	Nisling River	Donjek	Kluane River
Spawning areas	Chinook spawning below Onion Creek [p.4]		
Salmon observation/ Fishing sites	Chinook observed at mouth of Onion creek in early Sept. already spawned out [p.3]. Chinook observed in lower Nisling [p.3]	Salmon tag found at mouth of the Donjek [p.2] Chinook just above the mouth of Dip Creek on the Klotassin River (approx 30)[p.4]	Chum-Mile 1118 Swede Johnson Salmon Patch [p.1]
Fishing stories			“ It seems funny too, the bears they seem to really target them ones with (fisheries) tags. I dunno if they can just hook their claws into that tag and yank it out or what it is. They get a lot of them.” [p.11]
Recommendations			Prohibit ATV's from crossing spawning grounds, creeks--perhaps signage [p.10]

Kirk Johnson Burwash Landing	Tincup Lake/Creek	Kluane River	Kluane Lake
Spawning areas		Swede Johnson [p.7]	
Salmon observation/ Fishing sites	Chinook salmon [p.8]	Years ago used salmon for trapping bait [p.7]	Chinook 10 years ago below Burwash [p.5 & Spring Creek p.6] Chinook caught at Burwash Point [p.5]
Fishing Methods		Clubbing [p.14]	
Areas of open water/ spring water			Cultus Bay/ Groundwater [p.3]
Fishing stories		Salmon, “ a lot more, yeah. They used to run up the creek. Used to club them there. Still do that. I don’t even see how I never got pneumonia doing that. We used to run in the water and everything.” [p.14,15]	

Lena Johnson Burwash Landing	Koidern River	Tincup Lake/Creek	Donjek	Kluane River	Kluane Lake
Salmon observation/ Fishing sites	Chum salmon observed by Bessie John [p.10]...."She said salmon used to come up in some kind of water that ran on the grass, like through the field, swampy field and she said you know what? She said the kids got two salmon that they found there. She said maybe they're coming.." [p.10]	Family camp [p.1] Chinook..."my grandparents used to bring in a bundle of king salmon, ice salmon." [p.11] Grandparents dried salmon at camp [p.12]	White River First Nation Fishwheel nr Donjek and Kluane R. [p.11]	Chum--Salmon Patch [p.8] Chum used for trapping bait and dog feed [p.8&9] Observed tags on Chum [p.14]	Chinook caught in July by Lena's Mother on point below JoJo's house nr Burwash [p.2] Annie Hagen set nets for salmon nr Spring Creek [p.3] 'Not too long ago somebody saw 3-4 [chinook] of them going up the river nr Spring Creek [p.3] Last fall harvested 6 chum in lake [p.4]
Fishing Methods				Gaffing [p.8] Netting [p.3]	Nets [p.3] Nets [p.4]

Lena Johnson	Koidern R.	Tincup lake/creek	Donjek R.	Kluane River	Kluane lake
Areas of open water/ spring water					Spring water at Christmas Creek [p.4] Spring water at Cultus Bay [p.4] Open water at Cultus Point where the winter road goes across the Lake [p.5]
Fishing stories				(Chum Fishing) “They’d build like a box made out of logs, they fill edge with a bunch of little logs like that, and they build box on top of it with logs about this high and then that’s where they store the fish in Sept.”...”they gaff them and then put it in there” [p.8] Grandmother said”the salmon always been there. Whoever trapped before my dad. And Dorothy’s dad Sam. Yeah the story said the salmon always there.” [p.13]	

Mary Ann and Glenn Cottonwood Park	Kluane Lake
Salmon observation/ Fishing sites	No knowledge of chinook salmon in Kluane lake
Fishing Methods	Angling
Areas of open water/ Spring water	Open areas of water in the winter are around Cultus Bay, Thorsen Bay, Big Arm

Mary Easterson Burwash Landing	Tincup Lake/Creek	Kluane River
Spawning areas		
Salmon observation/ Fishing sites	Chinook [p.4] Family camps [p.4]	Glacier Creek Chum [p.1] Kluane River Chum [p.2] Swede Johnson Chum [p.2] Fish for dog food [p.2]

Ollie Wirth Resident 20 yrs Burwash Landing	Kluane Lake
Spawning areas	Chum--Christmas Creek [p.3]
Salmon observation/ Fishing sites	No one has ever caught a Chinook in their nets [p.3] Chum salmon have been caught [p.3]
Areas of open water/ spring water	Up the Arm there are underground springs bubbling up [p.7]

Peter Johnson Burwash Landing	Nisling River	Tincup Lake/Creek	Kluane River	Kluane Lake
Spawning areas			Glacier Creek [p.2]	Half breed Creek [p.2]
Salmon observation/ Fishing sites	Old fish camps [p.17]	Chinook observed/harvested [p.3] Family (mother) fished and dried salmon over a month [p.4]	Glacier Creek observed 5 tagged fish [p.5] Salmon Patch [p.5]	Two chinook harvested nr Burwash landing early 90's [p.2]
Fishing Methods		Traps [p.4, 15]		
Areas of open water/ spring water				Half Breed stays open [p.9] Close to Spring Creek nr Buck Dickson's place is open water [p.10] Narrows on the Little Arm open [p.10] Thorsen Bay "spring water I guess" [p.10] Raft Creek (Brooks Creek) up Talbot Arm [p.11]

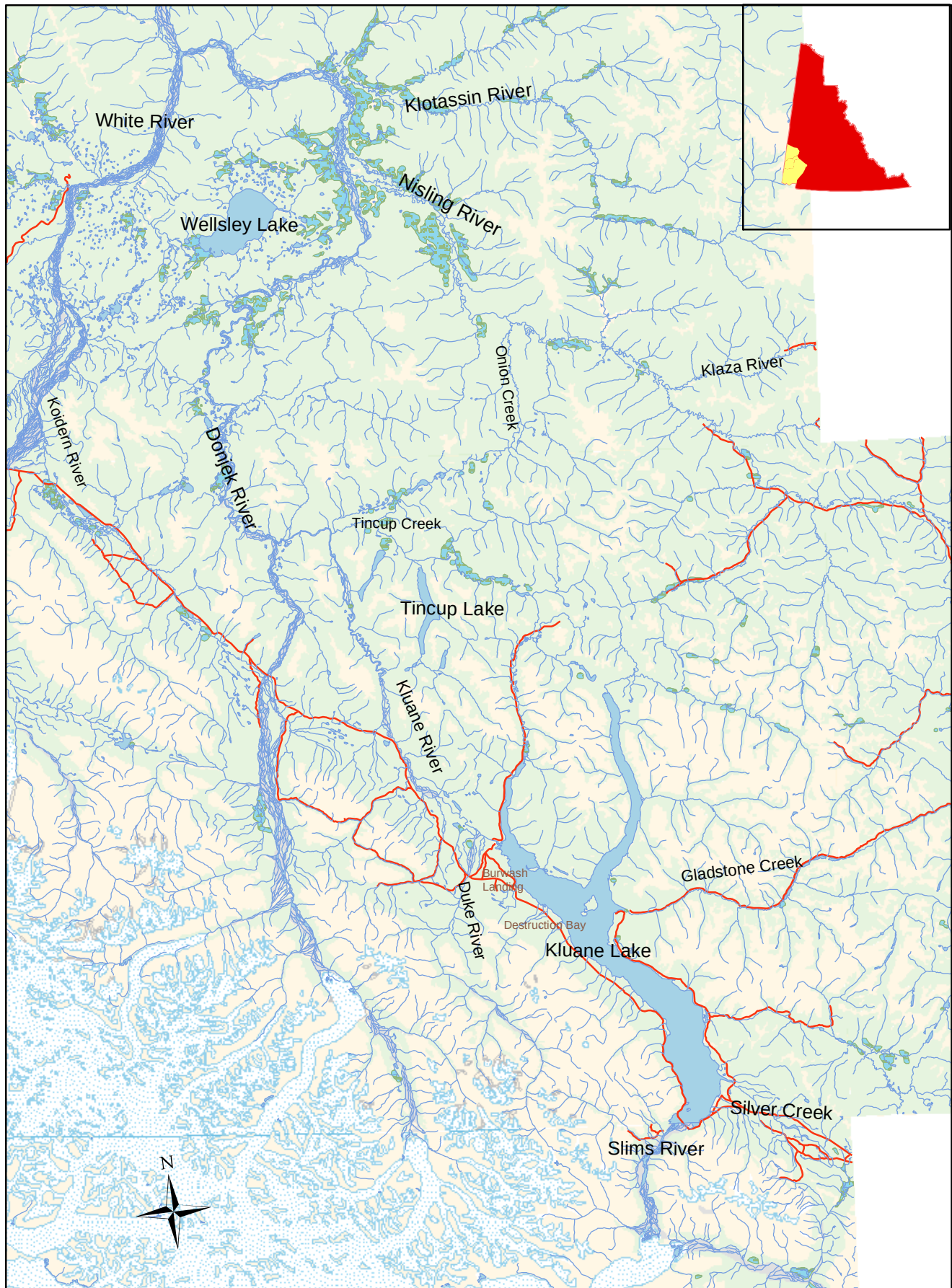
Ron Chambers Burwash Landing	Tincup Lake/Creek	Kluane River	Kluane Lake
Spawning areas		Salmon Patch chum [p.2] Swede Johnson chum [p.2]	Cultus Bay-chum [p.1]
Salmon observation/ Fishing sites	Chinook [p.3] Family camp [p.12]	Salmon Patch chum [p.2] Swede Johnson chum [p.2] Salmon for dog food [p.13] “well in salmon patch there is a good supply of grayling in there and big ones too.” [p.18]	Cultus Bay-chum [p.1] Silver Creek-chum [p.1] Christmas Creek-chum [p.1] Gladstone River-chum [p.1] Halfbreed Creek-chinook [p.2]
Fishing Methods	Fish traps [p.13]	Nets [p.2]	Nets [p.1]
Areas of open water/ spring water			“no no, not on Kluane Lake. The only place it stays thin is right in the narrows here at Brook’s Arm...even where streams come into the lake it’s a pretty solid freezing lake.”[p.2] “No, Spring Creek is open all year too and in here where the river comes out sometimes there’s some open water in there.” [p.3] Silver Creek-groundwater [p.3] No Name Creek-groundwater [p.3]

Ron Chambers	Tincup Lake/Creek	Kluane River	Kluane lake
Fishing stories		Salmon patch is pretty stable because that's a fair size spring creek that runs there. All you got is channel change ,right?...they've always been there." [p.2] "...is the fox farm at the lower end of the lake. They had a fox farm. Plus they had one at Silver City too. So they would have used fish exclusively fish for their fox farm" [p.13]	

Sharon Kabanak Burwash Landing	Kluane River	Kluane Lake
Spawning areas		The Bay nr Burwash landing [p.2]
Salmon observation/ Fishing sites	Family still fish for chum in the fall [p.8] Observes tags [p.9]	Set nets for Chinook for summer potlatch, just down from Burwash [p.4]
Fishing Methods	Gaffing [p.9]	Nets [p.4]
Areas of open water/ spring water		Where the Gladstone comes out [p.2] Little Arm [p.3]
Fishing stories	“My Mom used to catch dinner. It was October and she’d go down to the lake and set nets here. She would catch those chum salmon in the net in the bay here. So we would cook it a different way and...not really a survival food. It was just food that we ate. It tasted different. It tasted like water and stuff like some of them in the fall.”[p.9]	

MAPS

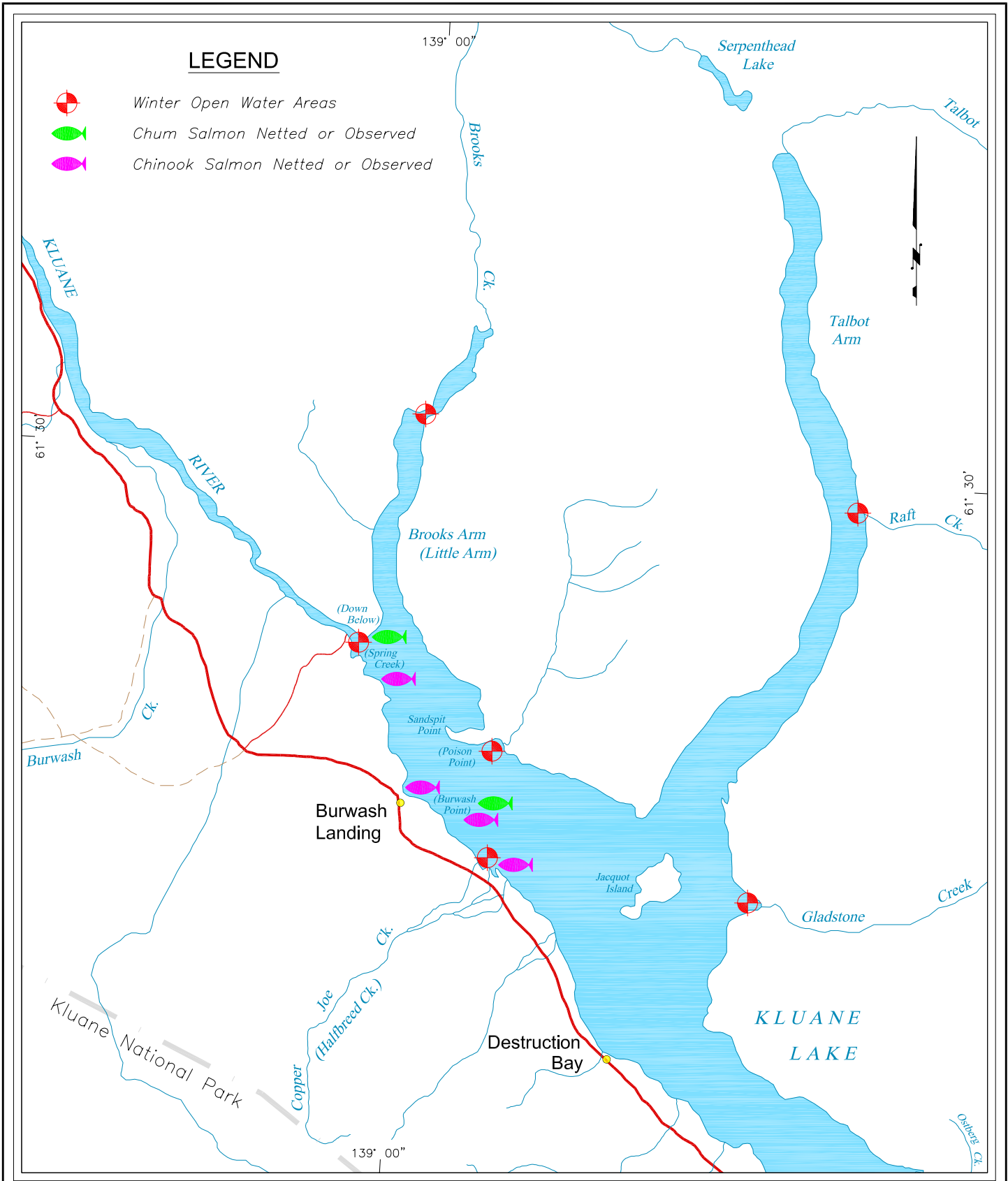
Fig.1 Kluane Region Watersheds



Scale
1:900,000



Data Source: NTDB250
NAD 83, UTM Zone 7
Map produced by Kluane First Nation.
Not to be used without permission of KFN.



KLUANE LAKE (North End)
WINTER OPEN WATER AREAS and SALMON OBSERVATIONS
 FIGURE 2A

